

# **ARP Monitor**

## **User Manual**

Real-Time ARP Spoofing Detection and LAN Intrusion Monitoring System

A beginner-friendly guide to installing, running, and using ARP Monitor.

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# 1. What is ARP Monitor?

ARP Monitor is a desktop application that watches your local network (LAN) for a type of attack called **ARP spoofing**. In ARP spoofing, an attacker tricks devices on the network into sending traffic to the attacker's computer instead of the real destination, often by pretending to be the network's router (gateway).

ARP Monitor listens to network traffic, keeps track of which IP address belongs to which device (MAC address), and raises an alert the moment something suspicious happens - for example, if a device suddenly claims to be your router.

**Important:** ARP Monitor only listens. It never sends fake network packets itself.

## 2. Installation

### Step 1 - Install Python

Make sure Python 3.10 or newer is installed on your computer.

### Step 2 - Install packet capture support

Windows users: install **Npcap** from [npcap.com](https://npcap.com) (choose the option "WinPcap API-compatible mode" during setup).

Linux/macOS users: packet capture support (libpcap) is usually already installed.

### Step 3 - Install the required Python packages

Open a terminal in the ARP\_Monitor folder and run:

```
pip install -r requirements.txt
```

## 3. How to Start the Application

Because ARP Monitor needs to read raw network traffic, it must be started with elevated privileges:

- **Windows:** Right-click your terminal (Command Prompt or PowerShell) and choose "Run as administrator", then run: `python main.py`
- **Linux / macOS:** Run: `sudo python3 main.py`

When the app opens, it automatically detects your network adapter, IP address, and router (gateway), and performs an initial scan to find devices already on the network. You'll see this information fill in on the Dashboard within a few seconds.

## 4. How to Monitor Your Network

Once the Dashboard has finished its initial scan, click the **Start Monitoring** button at the top of the Dashboard tab. This begins live packet capture in the background - the rest of the app stays fully responsive while it runs.

While monitoring is active, you can explore these tabs:

- **Devices** - every device discovered on your network, with vendor and authorization status.

- **Live Monitor** - a real-time table of every ARP packet as it's captured.
- **Alerts** - every suspicious event detected so far, filterable by severity.
- **Graphs** - visual charts summarizing alerts and device activity.

Tip: if you have devices you know are safe (your own phone, laptop, printer, etc.), select them in the Devices tab and click **Authorize Selected**. Authorized devices won't trigger "unknown device" alerts.

## 5. How to Read Alerts

Every alert has a severity level, shown with a color:

| Color  | Severity | Meaning                                                                                  |
|--------|----------|------------------------------------------------------------------------------------------|
| Green  | Info     | Something worth noting, e.g. a new device joined the network.                            |
| Yellow | Warning  | Unusual activity that may or may not be malicious - keep an eye on it.                   |
| Red    | Critical | A strong indicator of an active attack, e.g. a fake gateway or repeated MAC address chan |

When a new alert fires, a small popup appears in the corner of the screen, and critical alerts also play a sound and flash a warning indicator on the Dashboard. Every alert is also saved permanently and can be reviewed anytime in the Alerts tab.

If you receive a **CRITICAL - FAKE\_GATEWAY** alert, disconnect from the network and investigate immediately - this usually means someone is actively trying to intercept traffic.

## 6. Exporting Reports

Go to the **Reports** tab to generate a summary of your monitoring session. Three formats are available:

- **PDF** - a full report with tables and charts, ideal for sharing or printing.
- **CSV** - a spreadsheet-friendly list of all alerts.
- **TXT** - a simple plain-text summary.

Reports are saved in the `reports/` folder inside the application directory. You can also enable automatic report export every time monitoring stops, from the Settings tab.

## 7. Common Problems and Solutions

### The app says scapy is not installed

Run: `pip install scapy`, and make sure Npcap (Windows) or libpcap (Linux/macOS) is installed.

### "Permission denied" when starting monitoring

Restart the app with administrator (Windows) or `sudo` (Linux/macOS) privileges - raw packet capture requires it.

### No devices show up after a scan

Double check you selected the right network adapter in Settings, and that your device is connected to the same network you're trying to monitor.

### No sound or desktop popups appear

Check that Alert Sound and Desktop Notifications are enabled in the Settings tab. Desktop popups additionally require the optional 'plyer' package.

## **The interface looks plain, not styled**

Install the optional 'ttkbootstrap' package for the full dark theme: `pip install ttkbootstrap`

***End of User Manual.***